

# Nitheesh Karthikeyan

nitheesh.k2024@vitstudent.ac.in | +91 9842485832 | github.com/nitheeshx86 | linkedin.com/in/nitheeshx86

Engineering student with hands-on experience in software development, data analytics, and client-facing technology delivery - passionate about designing innovative, data-driven solutions to solve real-world business problems.

## EDUCATION

### VELLORE INSTITUTE OF TECHNOLOGY

2024--2028 | Chennai, India  
B.Tech in Electronics & Communication Engineering

### SBOA SCHOOL & JR. COLLEGE

2010--2024 | Chennai, India  
Multiple times winner in **Model United Nations**; strong structured communication, negotiation & policy analysis.

## SKILL SET

### Product Analytics:

Funnel Analysis, A/B Testing, User Journey Mapping, DAU/MAU Analysis, CAC/LTV

### Analytic and Team Tools:

Google Analytics, Mixpanel, Tableau, Excel, Notion, Jira (Agile/Scrum)

### Programming:

C++, Python, C, Java

### Developer Tools:

Git, Docker, Next.js, React

## LANGUAGES

**English** — Fluent Public Speaker  
**Tamil** — Native  
**Hindi** — Professional Working  
**French** — Limited Proficiency

## LEADERSHIP

**UN Millennium Fellow 2025 (UNAI)**  
— Top 4% of 60,000+ applicants; led sustainability-driven programs across 3+ communities; 10+ workshops reaching 200+ students.

**Paadhai (Career Awareness)** — Monthly sessions bridging career gaps for underserved govt. school students; 150+ impacted.

**Zerodha Varsity Lead Coordinator - Chennai Edition** — Managed cross-functional teams for city-wide finance competition: 500+ participants, INR 3.5L+ prize pool; re-invited for two subsequent external engagements based on prior competence of execution.

## REFERENCES

### Pranay V. Jain

Alumni, EY Chennai  
Assurance Associate  
(Oct 2021 – May 2022)  
+91 6383 230 387

### Dr. Reena Monica P.

Former Dean and current Asst. Director, International Affairs, VIT  
reenamonica@vit.ac.in  
+91 98408 37624

## EXPERIENCE

### SAMSUNG RESEARCH INDIA - BENGALURU | Student Researcher (PRISM)

February 2026 -- September 2026 | Hybrid

- Selected for **Samsung PRISM** — competitive industry-academia program; one of only **6 students from VIT**, working directly under two research scientists at SRI-B.
- Conducting applied research on **Multi-Core Tensorization of LLMs via Tensor Train Decomposition** — compressing 7B+ parameter models with <10% performance degradation.
- Executing a structured, phase-gated research roadmap: literature survey of LLM & Tensor Network state-of-the-art → comparative algorithm study (MPS, MPO, SVD) → baseline implementation → **large-scale HPC cluster optimization** → technical publication drafting.
- Applying sensitivity analysis & hyper-parameter tuning for optimal bond dimensions across multi-core TT decomposition configurations on high-performance computing infrastructure.

### MEDXAI | Research Software Intern (Stipend)

October 2024 -- February 2026

- Primary **technological partner** for **20+ pharmaceutical clients** — full cycle from requirement gathering through iterative development, testing, and deployment, mirroring a **consulting engagement lifecycle**.
- Engineered **Drug Explorer**: SMILES-to-insight data pipeline (binding affinity, drug synergy); now **outsourced to and actively used by major pharmaceutical companies**.
- Contributed to a drug candidate passing **wet-lab validation** from computational predictions — translating software outputs into measurable real-world health impact.
- Operated within **Agile/Scrum** environments across the full SDLC — sprint planning, daily stand-ups, and iterative release cycles using Jira; collaborated with cross-functional global teams.
- Produced data-oriented reports, dashboards, and client documentation for multinational clients including OmniActive, translating complex technical outputs into clear, executive-ready insights.
- Collaborated with interdisciplinary teams across research, engineering, and healthcare — building stakeholder communication and requirements translation skills essential for consulting contexts.

### FREELANCE | Web Developer & UI/UX Designer

December 2025 -- Present

- Designed & developed user-centric digital products for **8+ clients** — full SDLC from discovery to deployment; improved usability & retention by iterating on behavioral analytics and data-driven design thinking.

## PROJECTS

- LithoMap — Clinical Research Data Visualization**: Engineered an end-to-end analytical platform for **Govt. Stanley Medical College** to encode, process, and visualize complex FTIR spectroscopy clinical data. Optimized the data pipeline to support daily medical research workflows, successfully transitioning the application into active production by early 2026.
- S.P.L.A.S.H. — Smart Pothole Locator & Alert System**: Developed a vehicle-mounted edge-computing perception system for real-time road hazard identification. Trained custom computer vision models for anomaly detection and successfully tested the system as a functional MVP on a two-wheeler. Recognized among Top 3 Non-Startup Projects at iInventiv, IIT Madras.
- Quantum Federated ML for Fraud Detection (Qinnovision 2025, NYU Abu Dhabi)**: Investigated privacy-preserving anomaly detection architectures by applying quantum machine learning techniques to distributed financial datasets. Designed and simulated quantum circuits utilizing IBM Qiskit to evaluate the theoretical advantages of quantum federated learning in secure, decentralized environments.
- ML-Based Spatial Occupancy System**: Designed a real-time computer vision pipeline for spatial tracking and continuous availability inference. Deployed the system locally within the VIT campus library to optimize resource allocation through automated tracking. Awarded the **VP's Feather-in-the-Cap Award, VIT** for successful real-world algorithmic implementation.

## HACKATHONS

- IBM Qiskit Fall Fest 2024 — 1st Place**. Presented to senior leadership (Dr. L. Venkataraman) at IBM Quantum India; reviewed by international faculty.
- IBM Qiskit Fall Fest 2025 — Judge & Teaching Instructor**. Delivered sessions to **100+ participants** from 10+ countries; facilitated live mentoring.
- Intel + Microsoft SAS Hackathon (HEC Montreal) — World Rank 19**. Quantitative user-flow analysis to optimize business outcomes.
- Goldman Sachs Quantitative Finance Hackathon — AIR 163**. Alpha-generating strategies via portfolio analytics & quantitative modelling.

## AWARDS

- UN Millennium Fellow (2025)** — Top 4% of 60,000+ applicants; selected for global academic & social impact through technology.
- 1st Place — Best Startup Idea (2025)** — Presented by Dr. Soumya Swaminathan, former Chief Scientist, WHO.
- Nokia Campus Connect** — 1 of 5 from VIT Chennai; training in communication systems, ML & CS.
- VP's Feather-in-the-Cap Award, VIT** — ML-based Seat Occupancy System deployed in VIT Library.